

LAKSHMI BHARGAVI

Full Stack Engineer | React, Node.js, PostgreSQL, AI Applications

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PROFESSIONAL SUMMARY

Full Stack Engineer with 4+ years building production retail and payments platforms with **React, TypeScript, Node.js, PostgreSQL and GCP**. Delivered payments, 2FA, RBAC and third-party integrations for live retail platforms. Built RetailOps AI, a RAG-based retail assistant with LLM tool calling and LangGraph.

TECHNICAL SKILLS

Frontend:	React, Next.js, TypeScript, JavaScript, MUI, Tailwind CSS, React Hook Form, i18next
Backend and Data:	Node.js, Express.js, REST APIs, PostgreSQL, Sequelize, Datastore, OpenAPI
Cloud and DevOps:	GCP, App Engine, Secret Manager, Firebase, Bitbucket Pipelines, CI/CD
Security:	Authentication, Authorization, JWT, RBAC, TOTP/2FA, Encryption, secure cookies, CSRF
Testing and Delivery:	Vitest, React Testing Library, Node.js test runner, integration testing, Git, Agile
AI and Python:	Python, FastAPI, LLM APIs, RAG, tool calling, LangGraph, MCP

PROFESSIONAL EXPERIENCE

Full Stack Developer | Markovate Jul 2022 – Oct 2026

ROLE AND RESPONSIBILITIES

Owned full-stack delivery across three retail platforms, from React/TypeScript UIs and Node.js REST APIs to database design, deployment and production support, with automated testing and CI/CD.

CLIENT PROJECTS

Worked across three client projects, some in parallel.

Self-Checkout POS | Helius Retail Nov 2025 – Oct 2026

React / TypeScript frontend · Node.js / Express.js backend · Payment terminals

- Supported checkout across **50 stores** handling about **1,000 transactions per store per day**.
- Rebuilt the **React and TypeScript** kiosk around a secure backend, removing browser-accessible credentials and pricing logic to protect unattended checkout terminals.
- Built **Node.js payment orchestration** with adaptive polling and bounded retries to handle declined and pending payments safely and prevent repeat payment attempts.
- Delivered English/French scanning, checkout and receipts, with secure kiosk recovery after restarts, covered by automated frontend and backend tests.

Inventory Management | Helius Retail

Jun 2026 – Oct 2026

Next.js / React · TypeScript · PostgreSQL / Sequelize · Firebase

- Delivered **TOTP/2FA and RBAC** across frontend and backend, with encrypted secrets and store-scoped permissions to protect accounts and restrict access to authorized locations.
- Implemented **PostgreSQL** checkout transactions and row-level locks, enforcing promotion limits and ensuring stock and sales stay consistent under concurrent checkouts.
- Built stock adjustments that save all changes together, preventing partial updates; added automatic purchase-order suggestions and database locks to prevent duplicate restock drafts.
- Implemented sales reporting and profit-margin statistics using price and cost snapshots; fixed export and timezone issues to preserve accurate historical results.

Cashierless Retail | Aisle 24

Jul 2022 – Apr 2026

Node.js / Express.js · Cloud Datastore · GCP App Engine · Payments

- Delivered **6 core integrations**: Stripe, ProPay, Yoti, smart doors, coolers and lockers, enabling payments, identity and age verification, and automated access and purchasing in unattended stores.
- Implemented **Stripe PaymentIntents**, **Stripe Connect** payouts and **Global Payments/ProPay** split payments, supporting pre-authorization, franchise payouts and settlement-aware refunds.
- Built merchant onboarding and admin TOTP/2FA; extended GCP App Engine deployments, Bitbucket Pipelines and database indexes, supporting live operations across environments.

PROJECTS

RetailOps AI | AI-Enabled Retail Operations

Personal project, built independently. github.com/BhargaviAlimilli/RetailOps-AI

React / Next.js · Python / FastAPI · PostgreSQL · LLM APIs · LangGraph · MCP

- Built a retail assistant that queries **PostgreSQL** sales and inventory data, retrieves business documents and explains findings with cited sources through **RAG** and validated tool calling.
- Implemented **hybrid retrieval** using BM25 and vector search, rank fusion and reranking; added evaluation checks to compare retrieval relevance and answer groundedness.
- Created a **LangGraph** purchase-order workflow with saved state and human approval; exposed inventory tools through **MCP**.
- Used AI coding assistants during development and reviewed all generated code; added tests, logging and CI/CD checks; deployed on GCP.

EARLIER EXPERIENCE

Test Engineer | Marquios Technologies

Feb 2020 – Jun 2021

- Tested Android software releases, flashed firmware and reproduced defects, giving developers clear reports for troubleshooting and release verification.

EDUCATION

B.Tech | Electronics and Communication Engineering

2019

Jawaharlal Nehru Technological University, Kakinada